

Overview

Join us on our journey to design Australia's first lunar rover! The Australian Space Agency is supporting NASA's Artemis program with an Australian rover to collect samples from the Moon. The Big Dipper Lunar Regolith Acquisition Challenge allows YOU to be part of this mission. Phase 1 invites you to harness the power of Computer-Aided Design (CAD) to craft a Regolith Sample Acquisition Device, a pivotal component of this lunar rover. Phase 2 will give you the opportunity to build on designs by developing a list of Design Recommendations to inform and inspire future rover designs.

The Challenge

Imagine a lunar rover perched upon the Moon's surface, tasked with the objective of gathering and transporting lunar regolith to be used to extract oxygen. This mission will help pave the way for sustained human presence and exploration on the Moon and beyond. In this Phase 1 challenge, your mission is to design a Regolith Sample Acquisition Device that can be attached to an Australian designed rover for the collection of lunar soil (regolith) and deposit at an In-situ Resource Utilisation (ISRU) facility run by NASA. Phase 2 will provide the opportunity to integrate what is learnt from feedback and testing of Phase 1 winning designs into a set of design recommendations that will be useful for implementation.

Phase 1

Technical Background

The ISRU facility, a cornerstone of NASA's Artemis program, seeks to unlock the potential of lunar resources for propellant and life support systems. This initiative aligns with NASA's broader goal of establishing a sustainable human presence on the Moon and Mars. Aiming to be a global leader in exploration foundation services, the Australian Space Agency's Trailblazer program collaborates with NASA to contribute regolith samples to the ISRU facility, showcasing Australia's capabilities in resource mobility and remote operations.

The Rover

Your Solution will support the overall objectives of the rover. Key aspects of the rover which you will have to keep in mind when designing your system are included below:

The maximum mass of the rover is 20kg and the maximum stowed size is 50 x 40 x 40 cm. A suitable regolith acquisition system will have to fit on a rover of this size.

The rover may have to travel up to 100 meters to collect the regolith, and then transport it back to the NASA ISRU facility to deposit it. The acquisition system must not affect the rover's ability to travel these distances, with or without regolith on board. You may design a regolith acquisition system which collects any amount of regolith - maybe you choose to collect a large amount of regolith that potentially compromises the stability of the rover or maybe you choose to collect a small amount of regolith that might force the rover to make many trips back and forth. The options are vast, and the solutions are many. The rover will be solar powered. Due to its small size, it can only generate a small amount of power which it will have to share between all its subsystems. A design which requires an excessive amount of electrical power to operate will not be suitable.

The Regolith Acquisition System

Your task is to design a regolith acquisition system which will ride on-board the rover. A successful acquisition system will be:

Durable. The system cannot wear out after a few uses because no one will be on the Moon to service it. The system will also have to survive the violent vibration and shaking when it launches on top of a rocket.

Repeatable. The rover will have to complete back and forth trips to collect regolith (the exact number of trips will depend on your acquisition system size). The system will have to be consistent in collecting and depositing regolith.

Scalable. The mission intends to pave the way for future exploration of the Moon. A new, scaled up system, will be built after this initial demonstration to deliver greater quantities of regolith to an ISRU pilot plant.

Lunar Challenges

There are a number of challenges present in designing something for the Moon of which your design will have to take into account:

Lunar dust. The regolith, while valuable for its resources, is electrically charged and very fine. This means that it sticks to many materials and finds its way into cracks and gaps. Designs with moving components will need to be careful to ensure that joints are not eroded by the dust.

Gravity. The Moon is much smaller than Earth and so on the lunar surface, the gravitational field strength is 1/6 of that on the Earth's surface. The rover can be shifted quite easily if the acquisition has to lift, dig, push, or pull with great force. This presents

a risk that the rover may tip over or be damaged from the use of the acquisition system, which should be taken into account in the design.

Temperature. The temperature on the lunar surface in the polar regions can vary wildly from 40 degrees Celsius, down to -230 degrees Celsius in the shadows. This may rule out some choices of materials that cannot survive these temperature swings or ranges. Additionally, intricate mechanisms may not be able to function at these temperatures due to thermal expansion or contraction. A successful solution will need to be able to operate at these temperatures and anywhere in between.

As you can see, there are many considerations you need to make when designing things for the Moon. Are you brave enough to take this challenge on and try your hand at space engineering?

Communicating your design

The most important thing you can contribute to this challenge is your idea. However, a good idea needs to be communicated effectively. Your submission will include a CAD design to illustrate your concept, answers to the Submission Form questions to concisely describe how it works and how it meets the challenge requirements, and a short video explaining your design in simple terms.

Be part of Australia's contribution to a sustainable human presence on the Moon

Whether you are a seasoned CAD expert or someone who wants to pick up this skill, this challenge invites you to be part of an exciting mission. Your design could redefine how we acquire and utilise lunar regolith, paving the way for future astronauts.

Break Ground on Lunar Innovation

Are you ready to push the boundaries of lunar exploration? The Big Dipper Challenge is your entry into space. Step forward as an architect of the future, where the Moon's resources become assets that fuel our exploration beyond Earth. Your design holds the potential to redefine our relationship with the Moon and set the stage for humanity's next giant leap. Start designing today!

Challenge Timeline

Timeline

- Pre-registration: 2 October 2023
- Challenge Launch: 17 October 2023
- Submission Deadline: 18 December 2023 11:59pm AEDT

- Judging and Evaluation: 18 December 2023 – 5 January 2024
- Winners Announcement: 8 January 2024